

Program 9

Organize Kubernetes Pods Using Labels and Namespaces

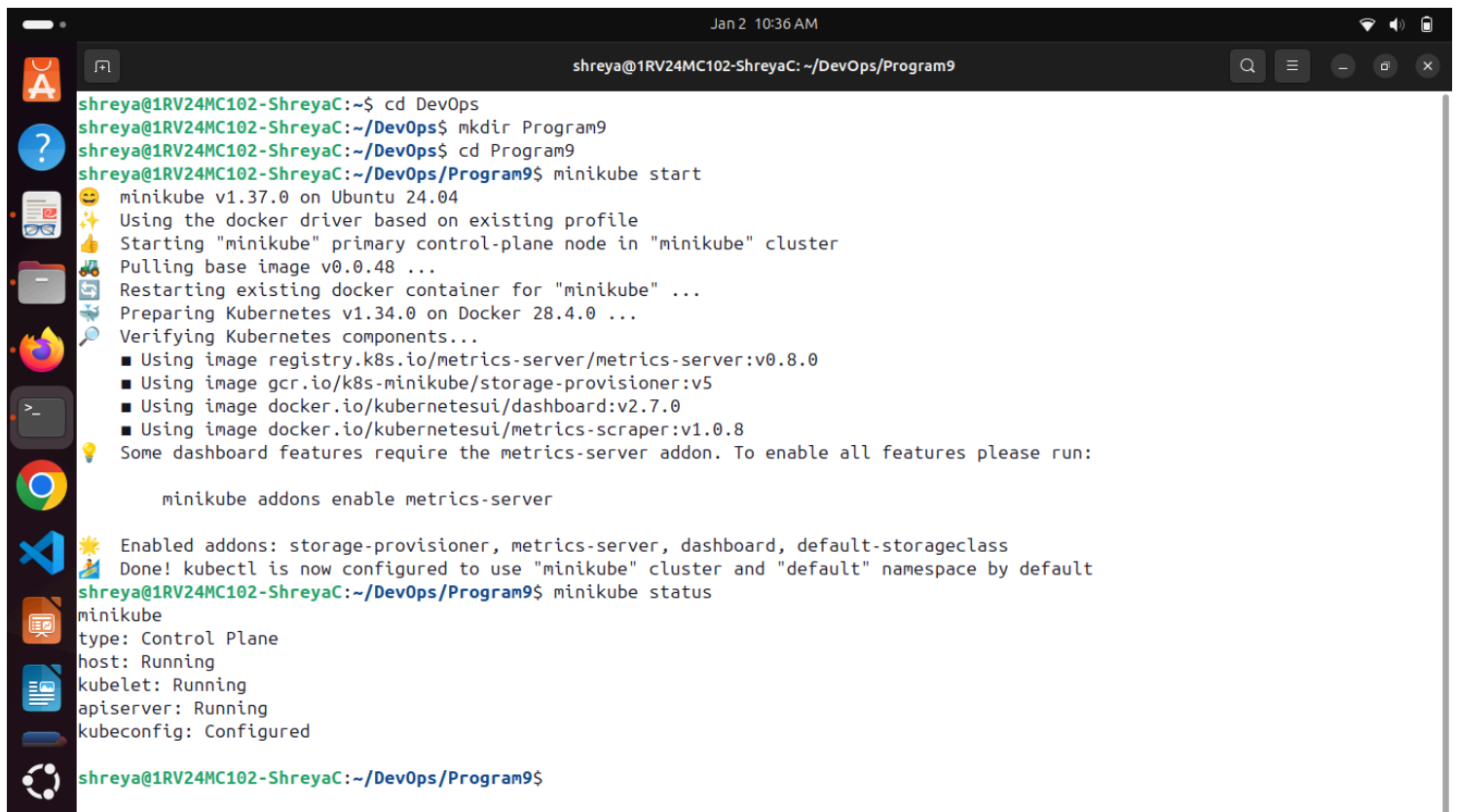
DevOps

— Program9

— namespaces.yaml

— pods.yaml

Step 1 : Create the folder for Program9 and start the minikube and check it's status.

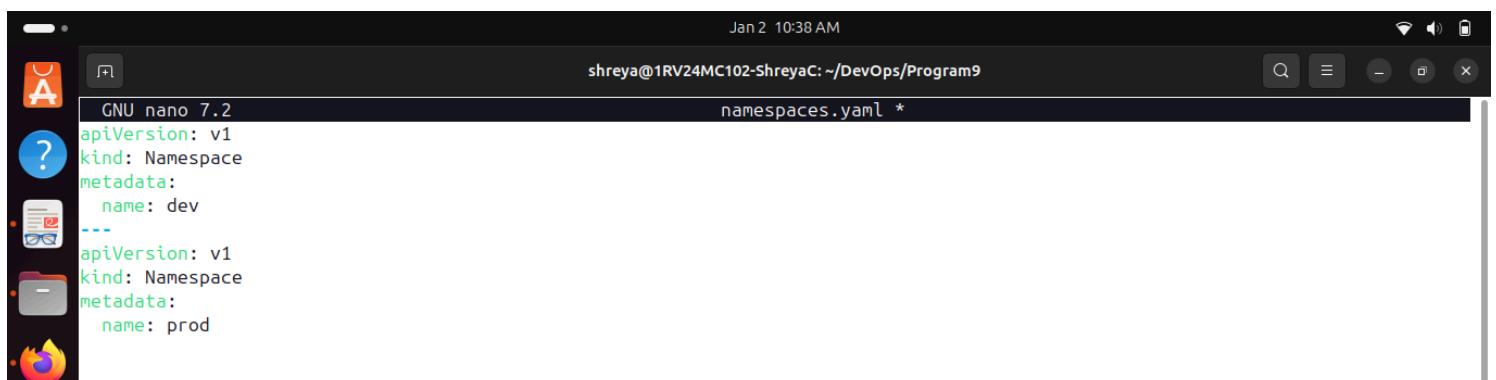


```
shreya@1RV24MC102-ShreyaC: ~/DevOps/Program9
shreya@1RV24MC102-ShreyaC:~$ cd DevOps
shreya@1RV24MC102-ShreyaC:~/DevOps$ mkdir Program9
shreya@1RV24MC102-ShreyaC:~/DevOps$ cd Program9
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ minikube start
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

  ■ Enabled addons: storage-provisioner, metrics-server, dashboard, default-storageclass
  ■ Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$
```

Step 2 : Add the following code in namespaces.yaml file



```
GNU nano 7.2 namespaces.yaml *
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 3 : Add the following code in pods.yaml file



```
GNU nano 7.2 pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step 4 : Apply the files



```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl apply -f pods.yaml
pod/frontend-pod unchanged
pod/backend-pod configured
pod/frontend-pod unchanged
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$
```

Step 6 : Verify the pods with namespaces

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   3 (37h ago) 15d
frontend-pod  1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   3 (37h ago) 15d
frontend-pod  1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$
```

Step 7 : Get all the pods

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -A
NAMESPACE      NAME                                     READY   STATUS    RESTARTS   AGE
default        demo-cron-29455508-4wb77               0/1     Completed 0           5m55s
default        demo-cron-29455510-f8zsl               0/1     Completed 0           3m55s
default        demo-cron-29455512-nv476               0/1     Completed 0           115s
default        nginx-deployment-6f9664446b-pmjwm      1/1     Running   1 (37h ago) 37h
default        nginx-deployment-6f9664446b-vqgcm      1/1     Running   1 (37h ago) 37h
dev            backend-pod                             1/1     Running   3 (37h ago) 15d
dev            frontend-pod                             1/1     Running   3 (37h ago) 15d
ingress-nginx  ingress-nginx-admission-create-4kwdt   0/1     Completed 0           32d
ingress-nginx  ingress-nginx-admission-patch-cfrdx     0/1     Completed 0           32d
ingress-nginx  ingress-nginx-controller-9cc49f96f-dhmsg 1/1     Running   7 (7m51s ago) 32d
kube-system    coredns-66bc5c9577-kg62x              1/1     Running   11 (37h ago) 37d
kube-system    etcd-minikube                           1/1     Running   11 (37h ago) 37d
kube-system    kube-apiserver-minikube                 1/1     Running   11 (37h ago) 37d
kube-system    kube-controller-manager-minikube        1/1     Running   12 (37h ago) 37d
kube-system    kube-proxy-h46vn                        1/1     Running   11 (37h ago) 37d
kube-system    kube-scheduler-minikube                 1/1     Running   11 (37h ago) 37d
kube-system    metrics-server-85b7d694d7-s98ps        1/1     Running   9 (7m35s ago) 31d
kube-system    storage-provisioner                     1/1     Running   23 (7m35s ago) 37d
kubernetes-dashboard dashboard-metrics-scraper-77bf4d6c4c-kpj7w 1/1     Running   1 (37h ago) 38h
kubernetes-dashboard kubernetes-dashboard-855c9754f9-w5w8s    1/1     Running   2 (7m35s ago) 38h
prod          backend-pod                             1/1     Running   3 (37h ago) 15d
prod          frontend-pod                             1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$
```

Step 8 : Verify all the pods with labels

```
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n prod -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   3 (37h ago) 15d
shreya@1RV24MC102-ShreyaC:~/DevOps/Program9$
```